12: Inference for Linear Combinations

Stat 230 - Spring 2026

```
library(ggformula)
library(broom)
```

Overview

The `cars` data set loaded below contains information on used cars for sale, including their `Price`, `Mileage`, and `Make` (a categorical variable with six levels).

```
cars <- read.csv("https://aluby.github.io/stat230/data/Cars.csv",
stringsAsFactors = TRUE)
```

We'll proceed with the `car_lm2` model:

```
car_lm2 <- lm(log(Price) ~ Mileage * Make, data = cars)

tidy(car_lm2)
```

```
# A tibble: 12 × 5
   term                      estimate  std.error statistic       p.value
   <chr>                        <dbl>      <dbl>     <dbl>         <dbl>
 1 (Intercept)           10.1          0.0845      119.     0
 2 Mileage               -0.00000710   0.00000392   -1.81   0.0704
 3 MakeCadillac           0.647        0.106         6.13   0.00000000141
 4 MakeChevrolet         -0.263        0.0915       -2.88   0.00411
 5 MakePontiac           -0.182        0.0987       -1.84   0.0657
 6 MakeSAAB               0.356        0.104         3.42   0.000664
 7 MakeSaturn            -0.322        0.117        -2.75   0.00617
 8 Mileage:MakeCadillac   0.000000158  0.00000495    0.0319 0.975
 9 Mileage:MakeChevrolet -0.00000162   0.00000425   -0.382  0.703
10 Mileage:MakePontiac    0.00000202   0.00000462    0.438  0.662
11 Mileage:MakeSAAB      -0.0000000492 0.00000476   -0.0103 0.992
12 Mileage:MakeSaturn    -0.00000367   0.00000539   -0.681  0.496
```

Calculating CIs for linear combinations

To calculate a confidence interval for a linear combination of coefficients, we must first calculate the estimate and standard error of the linear combination.

The slope for Cadillacs is the sum of the `Mileage` coefficient (the slope for the baseline, Buick) and the `Mileage:MakeCadillac` interaction coefficient:

$$\hat{\beta}_{\text{Mileage}} + \hat{\beta}_{\text{Mileage:MakeCadillac}}$$

Its standard error uses the variances *and* the covariance of the two estimates:

$$SE = \sqrt{\widehat{Var}\left(\hat{\beta}_{\text{Mileage}}\right) + \widehat{Var}\left(\hat{\beta}_{\text{Mileage:MakeCadillac}}\right) + 2\,\widehat{Cov}\left(\hat{\beta}_{\text{Mileage}}, \hat{\beta}_{\text{Mileage:MakeCadillac}}\right)}$$

First, extract the coefficients and covariance matrix from the model object.

```
car_coefs <- coef(car_lm2)
car_vcov <- vcov(car_lm2)
```

Line 1

Extract the coefficients from the model object and store them in `car_coefs`.

Line 2

Extract the covariance matrix from the model object and store it in `car_vcov`.

Next, calculate the estimate of the slope for Cadillacs.

```
estimate <- car_coefs[2] + car_coefs[8]
estimate
```

Line 1

`car_coefs` is a vector, so we pull off the necessary coefficients using their position in square brackets. Here, `car_coefs[2]` pulls off the coefficient for `Mileage` and `car_coefs[8]` pulls off the coefficient for the interaction term `Mileage:MakeCadillac`.

```
     Mileage
-6.940914e-06
```

Now, calculate the standard error of the slope for Cadillacs.

```
se <- sqrt(car_vcov[2,2] + car_vcov[8,8] + 2*car_vcov[2,8])
se
```

Line 1

`car_vcov` is a matrix, so we pull off the necessary variances and covariances using their row and column positions in square brackets. Here, `car_vcov[2,2]` pulls off the variance for `Mileage`, `car_vcov[8,8]` pulls off the variance for the interaction term `Mileage:MakeCadillac`, and `car_vcov[2,8]` pulls off the covariance between these two coefficients.

```
[1] 3.029984e-06
```

Finally, use the estimate and standard error to calculate a 95% confidence interval for the slope of the Cadillac model. You can use the `qt()` function to find the appropriate critical value.

Q1. Calculate a 95% confidence interval for the slope of the Cadillac model.

```
t_star <- qt(0.975, df = df.residual(car_lm2))
t_star
```

```
[1] 1.962964
```

```
ci <- estimate + c(-1, 1) * t_star * se
ci
```

```
[1] -1.288866e-05 -9.931655e-07
```

The critical value is $t^* = 1.963$ (on 792 degrees of freedom). The estimated slope for Cadillacs is −6.94e-06 with a standard error of 3.03e-06, so the 95% confidence interval is (-1.29e-05, −9.93e-07).

Interpretation: the response is $\log(\text{Price})$, so we back-transform. We are 95% confident that, for Cadillacs, every additional 10,000 miles multiplies the expected price by a factor between 0.879 and 0.99, i.e., a decrease of about 1% to 12.1%.

Check: change the baseline

If we make Cadillac the baseline level of `Make`, the `Mileage` coefficient *is* the Cadillac slope, so R reports its confidence interval directly:

```
cars_cad <- cars |>
  dplyr::mutate(Make = relevel(Make, ref = "Cadillac"))

car_lm_cad <- lm(log(Price) ~ Mileage * Make, data = cars_cad)
confint(car_lm_cad, "Mileage")
```

```
                2.5 %        97.5 %
Mileage -1.288866e-05 -9.931655e-07
```

This matches the interval above.